

ADVANCED

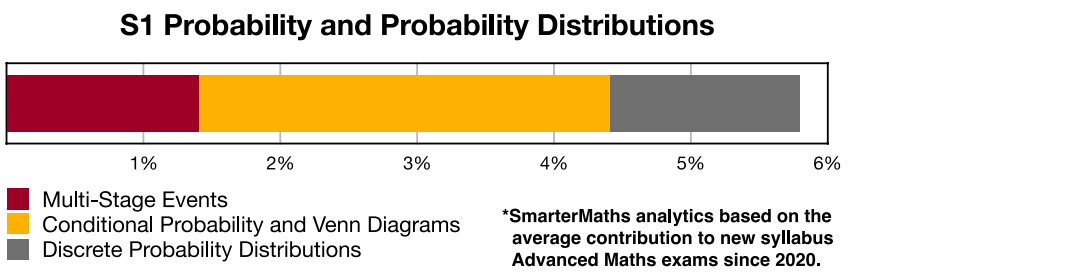
Stage 6

Statistics (Adv), S1 Probability & Probability Distributions (Adv)

Conditional Probability and Venn Diagrams (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 79.5 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- S1 Probability* has contributed an average of 5.8% per new syllabus Advanced exam since it was introduced in 2020.
- S1 Probability* has been split into four sub-categories for the purposes of this analysis: 1-*Multi-Stage Events* (1.4%), 2-*Relative Frequency* (0.0%), 3-*Conditional Probability and Venn Diagrams* (3.0%) and 4-*Discrete Probability Distributions* (1.4%).
- This analysis looks at *Conditional Probability and Venn Diagrams*.

HSC ANALYSIS - What to expect and common pitfalls

- Conditional Probability and Venn Diagrams* (3.0%) has been examined in 4 of the 5 new syllabus exams to date, including significant 5-mark allocations in 2023 and 2020.
- Conditional Probability* as a "concept" has been examined in past *Advanced* exams through probability trees and selection without replacement. Note that these question types are both covered under the separate topic *Multi-stage Events*.
- However, the new syllabus states explicitly that understanding Venn diagrams and the use of specific notation is now also required. A concerning knowledge gap was flagged by 2023 *Adv Q31* which used specific notation and was terribly answered. A "must review" question in all parts. Please pay specific attention to the concept of independent events in this context!
- Venn Diagrams can be extremely effective in solving conditional probability questions (review 2020 *Adv 14*). This important solution strategy appears in many of the worked solutions.

Questions

1. Probability, 2ADV S1 2024 HSC 2 MC

In a group of 60 students, 38 play basketball, 35 play hockey and 5 do not play either basketball or hockey.

How many students play both basketball and hockey?

- A. 55
- B. 18
- C. 13
- D. 8

2. Probability, 2ADV S1 2019 MET2 11 MC

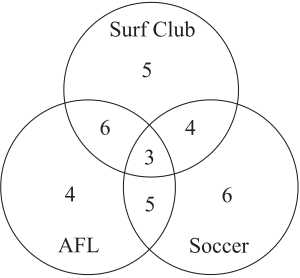
A and B are events from a sample space such that $P(A) = p$, where $p > 0$, $P(B | A) = m$ and $P(B | A') = n$.

A and B are independent events when

- A. $m = n$
- B. $m = 1 - p$
- C. $m + n = 1$
- D. $m = p$

3. Probability, 2ADV S1 SM-Bank 4 MC

In a classroom, students are asked what sports club they are members of and the results are shown in the Venn diagram.



A student who is a member of a soccer club is chosen at random. What is the probability that he/she is also a member of a surf club?

- A. $\frac{2}{5}$
- B. $\frac{4}{11}$
- C. $\frac{2}{9}$
- D. $\frac{7}{18}$

4. Probability, 2ADV S1 2012 MET2 13 MC

A and B are events of a sample space S .

$P(A \cap B) = \frac{2}{5}$ and $P(A \cap B') = \frac{3}{7}$

$P(B' \mid A)$ is equal to

- A. $\frac{6}{35}$
- B. $\frac{15}{29}$
- C. $\frac{14}{35}$
- D. $\frac{29}{35}$

5. Probability, 2ADV S1 2013 MET2 10 MC

For events A and B , $P(A \cap B) = p$, $P(A' \cap B) = p - \frac{1}{8}$ and $P(A \cap B') = \frac{3p}{5}$.

If A and B are independent, then the value of p is

- A. 0
- B. $\frac{1}{4}$
- C. $\frac{3}{8}$
- D. $\frac{1}{2}$

6. Probability, 2ADV S1 2010 MET2 21 MC

Events A and B are mutually exclusive events of a sample space with

$P(A) = p$ and $P(B) = q$ where $0 < p < 1$ and $0 < q < 1$.

$P(A' \cap B')$ is equal to

- A. $(1 - p)(1 - q)$
- B. $1 - pq$
- C. $1 - (p + q)$
- D. $1 - (p + q - pq)$

7. Probability, 2ADV S1 2013 MET2 17 MC

A and B are events of a sample space.

Given that $P(A \mid B) = p$, $P(B) = p^2$ and $P(A) = p^{\frac{1}{3}}$, $P(B \mid A)$ is equal to

- A. p^3
- B. $p^{\frac{4}{3}}$
- C. $p^{\frac{7}{3}}$
- D. $p^{\frac{8}{3}}$

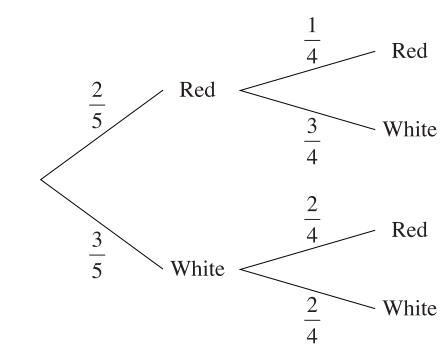
8. Probability, 2ADV S1 2014 MET2 14 MC

If X is a random variable such that $P(X > 5) = a$ and $P(X > 8) = b$, then $P(X < 5 \mid X < 8)$ is

- A. $\frac{a}{b}$
- B. $\frac{a - b}{1 - b}$
- C. $\frac{1 - b}{1 - a}$
- D. $\frac{a - 1}{b - 1}$

9. Probability, 2ADV S1 2024 HSC 9 MC

A bag contains 2 red and 3 white marbles. Jovan randomly selects two marbles at the same time from this bag. The probability tree diagram shows the probabilities for each of the outcomes.



Given that one of the marbles that Jovan has selected is red, what is the probability that the other marble that he has selected is also red?

- a. $\frac{1}{10}$
- b. $\frac{1}{7}$
- c. $\frac{1}{4}$
- d. $\frac{7}{10}$

10. Probability, 2ADV S1 2012 MET1 2

A car manufacturer is reviewing the performance of its car model X. It is known that at any given six-month service, the probability of model X requiring an oil change is $\frac{17}{20}$, the probability of model X requiring an air filter change is $\frac{3}{20}$ and the probability of model X requiring both is $\frac{1}{20}$.

- a. State the probability that at any given six-month service model X will require an air filter change without an oil change. (1 mark)
- b. The car manufacturer is developing a new model. The production goals are that the probability of model Y requiring an oil change at any given six-month service will be $\frac{m}{m+n}$, the probability of model Y requiring an air filter change will be $\frac{n}{m+n}$ and the probability of model Y requiring both will be $\frac{1}{m+n}$, where $m, n \in \mathbb{Z}^+$.

Determine m in terms of n if the probability of model Y requiring an air filter change without an oil change at any given six-month service is 0.05. (2 marks)

11. Probability, 2ADV S1 SM-Bank 3

- In a workplace of 25 employees, each employee speaks either French or German, or both.
- If 36% of the employees speak German, and 20% speak both French and German.
- i. Calculate the probability one person chosen could speak German if they could speak French. Give your answer to the nearest percent. (1 mark)
 - ii. Calculate the probability one person chosen could not speak French if they could speak German. Give your answer to the nearest percent. (1 mark)

12. Probability, 2ADV S1 2020 HSC 14

- History and Geography are two of the subjects students may decide to study. For a group of 40 students, the following is known.
- 7 students study neither History nor Geography
 - 20 students study History
 - 18 students study Geography
- a. A student is chosen at random. By using a Venn diagram, or otherwise, find the probability that the student studies both History and Geography. (2 marks)
 - b. A student is chosen at random. Given that the student studies Geography, what is the probability that the student does NOT study History? (1 mark)
 - c. Two different students are chosen at random, one after the other. What is the probability that the first student studies History and the second student does NOT study History? (2 marks)

13. Probability, 2ADV S1 2009 MET1 5

- Four identical balls are numbered 1, 2, 3 and 4 and put into a box. A ball is randomly drawn from the box, and not returned to the box. A second ball is then randomly drawn from the box.
- i. What is the probability that the first ball drawn is numbered 4 and the second ball drawn is numbered 1? (1 mark)
 - ii. What is the probability that the sum of the numbers on the two balls is 5? (1 mark)
 - iii. Given that the sum of the numbers on the two balls is 5, what is the probability that the second ball drawn is numbered 1? (2 marks)

14. Probability, 2ADV S1 2015 MET1 8

- For events A and B from a sample space, $P(A \mid B) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$.
- i. Calculate $P(A \cap B)$. (1 mark)
 - ii. Calculate $P(A^c \cap B)$, where A^c denotes the complement of A . (1 mark)
 - iii. If events A and B are independent, calculate $P(A \cup B)$. (1 mark)

15. Probability, 2ADV S1 2007 MET1 11

There is a daily flight from Paradise Island to Melbourne. The probability of the flight departing on time, given that there is fine weather on the island, is 0.8, and the probability of the flight departing on time, given that the weather on the island is not fine, is 0.6.

In March the probability of a day being fine is 0.4.

Find the probability that on a particular day in March

- i. the flight from Paradise Island departs on time (2 marks)
- ii. the weather is fine on Paradise Island, given that the flight departs on time. (2 marks)

16. Probability, 2ADV S1 2011 MET1 8

Two events, A and B , are such that $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{4}$.

If A' denotes the compliment of A , calculate $P(A' \cap B)$ when

- i. $P(A \cup B) = \frac{3}{4}$ (2 marks)
- ii. A and B are mutually exclusive. (1 mark)

17. Probability, 2ADV S1 2014 MET1 9

Sally aims to walk her dog, Mack, most mornings. If the weather is pleasant, the probability that she will walk Mack is $\frac{3}{4}$, and if the weather is unpleasant, the probability that she will walk Mack is $\frac{1}{3}$.

Assume that pleasant weather on any morning is independent of pleasant weather on any other morning.

- a. In a particular week, the weather was pleasant on Monday morning and unpleasant on Tuesday morning.
Find the probability that Sally walked Mack on at least one of these two mornings. (2 marks)
- b. In the month of April, the probability of pleasant weather in the morning was $\frac{5}{8}$.
 - i. Find the probability that on a particular morning in April, Sally walked Mack. (2 marks)
 - ii. Using your answer from **part b.i.**, or otherwise, find the probability that on a particular morning in April, the weather was pleasant, given that Sally walked Mack that morning. (2 marks)

18. Probability, 2ADV S1 2017 MET1 8

For events A and B from a sample space, $P(A|B) = \frac{1}{5}$ and $P(B|A) = \frac{1}{4}$. Let $P(A \cap B) = p$.

- i. Find $P(A)$ in terms of p . (1 mark)
- ii. Find $P(A' \cap B')$ in terms of p . (2 marks)
- iii. Given that $P(A \cup B) \leq \frac{1}{5}$, state the largest possible interval for p . (2 marks)

19. Probability, 2ADV S1 2022 HSC 15

In a bag there are 3 six-sided dice. Two of the dice have faces marked 1, 2, 3, 4, 5, 6. The other is a special die with faces marked 1, 2, 3, 5, 5, 5.

One die is randomly selected and tossed.

- a. What is the probability that the die shows a 5? (1 mark)
- b. Given that the die shows a 5, what is the probability that it is the special die? (1 mark)

20. Probability, 2ADV S1 2023 HSC 31

Four Year 12 students want to organise a graduation party. All four students have the same probability, $P(F)$, of being available next Friday. All four students have the same probability, $P(S)$, of being available next Saturday.

It is given that $P(F) = \frac{3}{10}$, $P(S | F) = \frac{1}{3}$, and $P(F | S) = \frac{1}{8}$.

Kim is one of the four students.

- a. Is Kim's availability next Friday independent from his availability next Saturday? Justify your answer. (1 mark)
- b. Show that the probability that Kim is available next Saturday is $\frac{4}{5}$. (2 marks)
- c. What is the probability that at least one of the four students is NOT available next Saturday? (2 marks)

21. Probability, 2ADV S1 2007 MET1 6

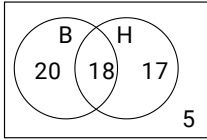
Two events, A and B , from a given event space, are such that $P(A) = \frac{1}{5}$ and $P(B) = \frac{1}{3}$.

- i. Calculate $P(A' \cap B)$ when $P(A \cap B) = \frac{1}{8}$. (1 mark)
- ii. Calculate $P(A' \cap B)$ when A and B are mutually exclusive events. (1 mark)

Worked Solutions

1. Probability, 2ADV S1 2024 HSC 2 MC

Method 1:



Method 2:

$$n(B \cup H) = n(B) + n(H) - n(B \cap H)$$

$$55 = 38 + 35 - n(B \cap H)$$

$$n(B \cap H) = 18$$

$$\Rightarrow B$$

2. Probability, 2ADV S1 2019 MET2 11 MC

Since A and B are independent:

$$P(B | A) = P(B | A')$$

$$\therefore m = n$$

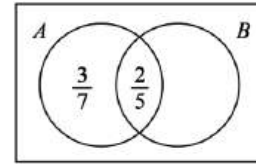
$$\Rightarrow A$$

3. Probability, 2ADV S1 SM-Bank 4 MC

$$\begin{aligned} P(\text{Surf} | \text{Soccer}) &= \frac{n(\text{Surf} \cap \text{Soccer})}{n(\text{Soccer})} \\ &= \frac{3 + 4}{3 + 4 + 5 + 6} \\ &= \frac{7}{18} \end{aligned}$$

$$\Rightarrow D$$

4. Probability, 2ADV S1 2012 MET2 13 MC



$$\begin{aligned} P(B' | A) &= \frac{P(B' \cap A)}{P(A)} \\ &= \frac{P(B' \cap A)}{P(B' \cap A) + P(A \cap B)} \\ &= \frac{\frac{3}{7}}{\frac{3}{7} + \frac{2}{5}} \\ &= \frac{15}{29} \\ \Rightarrow B \end{aligned}$$

5. Probability, 2ADV S1 2013 MET2 10 MC

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B') \\ &= p + \frac{3p}{5} \\ &= \frac{8p}{5} \end{aligned}$$

$$\begin{aligned} P(B) &= P(B \cap A) + P(B \cap A') \\ &= p + p - \frac{1}{8} \\ &= 2p - \frac{1}{8} \end{aligned}$$

Since A and B are independent events,

$$P(A \cap B) = P(A) \times P(B)$$

$$p = \frac{8p}{5} \left(2p - \frac{1}{8} \right)$$

$$5p = 16p^2 - p$$

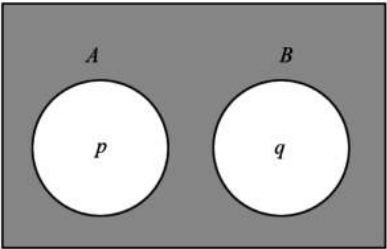
$$16p^2 - 6p = 0$$

$$2p(8p - 3) = 0$$

$$\therefore p = \frac{3}{8}, \quad p \neq 0$$

$$\Rightarrow C$$

6. Probability, 2ADV S1 2010 MET2 21 MC



$$\begin{aligned} P(A' \cap B') &= 1 - p - q \\ &= 1 - (p + q) \\ \Rightarrow C \end{aligned}$$

◆ Mean mark 43%.

7. Probability, 2ADV S1 2013 MET2 17 MC

$$\begin{aligned} P(A \mid B) &= \frac{P(A \cap B)}{P(B)} \\ p &= \frac{P(A \cap B)}{p^2} \\ \therefore P(A \cap B) &= p^3 \end{aligned}$$

◆ Mean mark 49%.

$$\begin{aligned} P(B \mid A) &= \frac{P(A \cap B)}{P(A)} \\ &= \frac{p^3}{p^{\frac{1}{3}}} \\ \therefore P(B \mid A) &= p^{\frac{8}{3}} \\ \Rightarrow D \end{aligned}$$

8. Probability, 2ADV S1 2014 MET2 14 MC

$$\begin{aligned} P(X < 5) \mid P(X < 8) &= \frac{P(X < 5 \cap X < 8)}{P(X < 8)} \\ &= \frac{P(X < 5)}{P(X < 8)} \\ &= \frac{1-a}{1-b} \\ &= \frac{a-1}{b-1} \\ \Rightarrow D \end{aligned}$$

◆ Mean mark 45%.

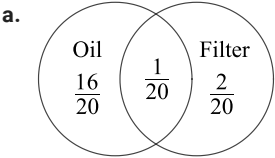
9. Probability, 2ADV S1 2024 HSC 9 MC

$$\begin{aligned} P(R_2 \mid \text{other is red}) &= \frac{P(R_1 \cap R_2)}{P(\text{at least 1 red})} \\ &= \frac{\frac{2}{5} \times \frac{1}{4}}{1 - P(WW)} \\ &= \frac{\frac{2}{5} \times \frac{1}{4}}{1 - (\frac{3}{5} \times \frac{2}{4})} \\ &= \frac{1}{7} \end{aligned}$$

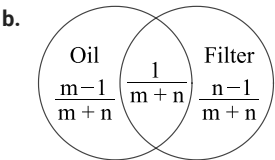
◆◆◆ Mean mark 14%.

⇒ B

10. Probability, 2ADV S1 2012 MET1 2



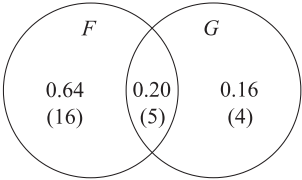
$$\begin{aligned} \Pr(F \cap O^c) &= \Pr(F) - \Pr(F \cap O) \\ &= \frac{3}{20} - \frac{1}{20} \\ &= \frac{1}{10} \end{aligned}$$



$$\begin{aligned} \Pr(F \cap O^c) &= \frac{n}{m+n} - \frac{1}{m+n} \\ \frac{1}{20} &= \frac{n-1}{m+n} \\ m+n &= 20n-20 \\ m &= 19n-20 \end{aligned}$$

11. Probability, 2ADV S1 SM-Bank 3

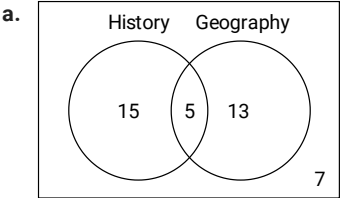
i. Expressing in a Venn diagram:



$$P(G \mid F) = \frac{P(G \cap F)}{P(F)}$$
$$= \frac{0.20}{0.84}$$
$$= 0.238 \dots$$
$$= 24\%$$

ii.
$$P(\text{not } F \mid G) = \frac{P(\text{not } F \cap G)}{P(G)}$$
$$= \frac{0.16}{0.36}$$
$$= 0.444 \dots$$
$$= 44\%$$

12. Probability, 2ADV S1 2020 HSC 14



$$P(\text{H and G}) = \frac{5}{40}$$
$$= \frac{1}{8}$$

b.
$$P(\overline{\text{H}} \mid \text{G}) = \frac{P(\overline{\text{H}} \cap \text{G})}{P(\text{G})}$$
$$= \frac{13}{18}$$

c.
$$P(\text{H}, \overline{\text{H}}) = \frac{20}{40} \times \frac{20}{39}$$
$$= \frac{10}{39}$$

♦ Mean mark (b) 49%.

13. Probability, 2ADV S1 2009 MET1 5

i.
$$P(4, 1) = \frac{1}{4} \times \frac{1}{3}$$
$$= \frac{1}{12}$$

ii.
$$P(\text{Sum} = 5)$$
$$= P(1, 4) + P(2, 3) + P(3, 2) + P(4, 1)$$
$$= 4 \times \left(\frac{1}{4} \times \frac{1}{3} \right)$$
$$= \frac{1}{3}$$

iii. Conditional Probability

$$P(2^{\text{nd}} = 1 \mid \text{Sum} = 5)$$
$$= \frac{P(4, 1)}{P(\text{Sum} = 5)}$$
$$= \frac{\frac{1}{12}}{\frac{1}{3}}$$
$$= \frac{1}{4}$$

♦ Mean mark 46%.

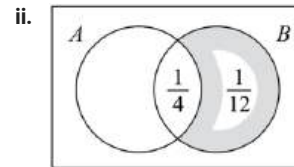
14. Probability, 2ADV S1 2015 MET1 8

i. Using Conditional Probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$\frac{3}{4} = \frac{P(A \cap B)}{\frac{1}{3}}$$

$$\therefore P(A \cap B) = \frac{1}{4}$$



$$P(A' \cap B) = P(B) - P(A \cap B)$$
$$= \frac{1}{3} - \frac{1}{4}$$
$$= \frac{1}{12}$$

iii. If A, B independent

$$P(A \cap B) = P(A) \times P(B)$$

$$\frac{1}{4} = P(A) \times \frac{1}{3}$$

$$\therefore P(A) = \frac{3}{4}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

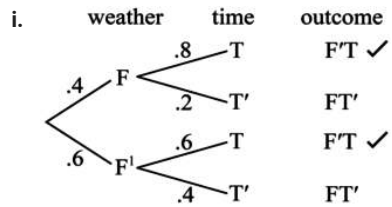
$$= \frac{3}{4} + \frac{1}{3} - \frac{1}{4}$$

$$\therefore P(A \cup B) = \frac{5}{6}$$

♦♦ Mean mark 28%.

MARKER'S COMMENT: A lack of understanding of independent events was clearly evident.

15. Probability, 2ADV S1 2007 MET1 11



$$\begin{aligned}
 P(FT) + P(F'T) \\
 &= 0.4 \times 0.8 + 0.6 \times 0.6 \\
 &= 0.32 + 0.36 \\
 &= 0.68
 \end{aligned}$$

ii. Conditional probability:

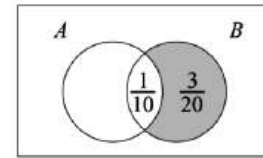
$$\begin{aligned}
 P(F | T) &= \frac{P(F \cap T)}{P(T)} \\
 &= \frac{0.32}{0.68}
 \end{aligned}$$

$$\therefore P(F | T) = \frac{8}{17}$$

♦♦ Mean mark 29%.
MARKER'S COMMENT: Students continue to struggle with conditional probability. Attention required here.

16. Probability, 2ADV S1 2011 MET1 8

i. Sketch Venn Diagram



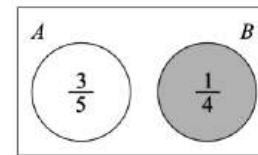
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{3}{4} = \frac{3}{5} + \frac{1}{4} - P(A \cap B)$$

$$P(A \cap B) = \frac{1}{10}$$

$$\therefore P(A' \cap B) = \frac{1}{4} - \frac{1}{10} = \frac{3}{20}$$

ii.



$$P(A \cap B) = 0 \text{ (mutually exclusive),}$$

$$\therefore P(A' \cap B) = P(B) = \frac{1}{4}$$

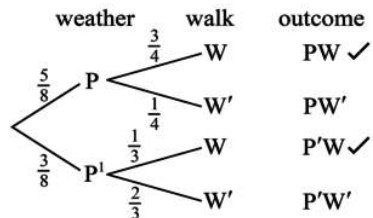
17. Probability, 2ADV S1 2014 MET1 9

a. $P(\text{at least 1 walk}) = 1 - P(\text{no walk})$

$$= 1 - \frac{1}{4} \times \frac{2}{3}$$

$$= \frac{5}{6}$$

b.i. Construct tree diagram:



$$P(PW) + P(P'W) = \frac{5}{8} \times \frac{3}{4} + \frac{3}{8} \times \frac{1}{3}$$

$$= \frac{19}{32}$$

b.ii. $P(P | W) = \frac{P(P \cap W)}{P(W)}$

$$= \frac{\frac{5}{8} \times \frac{3}{4}}{\frac{19}{32}}$$

$$= \frac{15}{32} \times \frac{32}{19}$$

$$= \frac{15}{19}$$

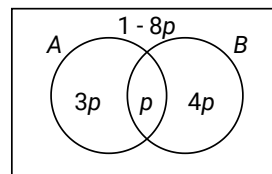
♦ Part (b)(ii) mean mark 38%.

18. Probability, 2ADV S1 2017 MET1 8

i. $P(A) = \frac{P(A \cap B)}{P(B|A)}$

$$= \frac{\frac{1}{4}}{4p}$$

ii. Consider the Venn diagram:



$$P(A \cap B) = 1 - 8p$$

iii. Given $P(A \cup B) = 8p$

$$\Rightarrow 0 < 8p \leq \frac{1}{5}$$

$$\therefore 0 < p \leq \frac{1}{40}$$

♦ Mean mark 40%.

MARKER'S COMMENT: The most successful answers used a Venn diagram or table.

♦ Mean mark 37%.

19. Probability, 2ADV S1 2022 HSC 15

a. $P(5) = \frac{1}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{1}{2}$

$$= \frac{1}{18} + \frac{1}{18} + \frac{1}{6}$$

$$= \frac{5}{18}$$

Mean mark (a) 53%.

♦ Mean mark (b) 40%.

b. $P(\text{special die given a 5 is rolled})$

$$= \frac{P(\text{special die}) \cap P(5)}{P(5)}$$

$$= \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{5}{18}}$$

$$= \frac{1}{6} \times \frac{18}{5}$$

$$= \frac{3}{5}$$

20. Probability, 2ADV S1 2023 HSC 31

a. $P(F) \neq P(F|S)$ which is not the case ($\frac{3}{10} \neq \frac{1}{8}$)
 $\therefore$ Kim's availability on Friday is not independent of Saturday

◆◆ Mean mark (a) 18%.

b.
$$P(S|F) = \frac{P(S) \cap P(F)}{P(F)}$$
$$\frac{1}{3} = \frac{P(S) \cap P(F)}{\frac{3}{10}}$$
$$\frac{1}{10} = P(S) \cap P(F)$$

◆ Mean mark (b) 44%.

$$P(F|S) = \frac{P(F) \cap P(S)}{P(S)}$$
$$\frac{1}{8} = \frac{\frac{1}{10}}{P(S)}$$
$$\frac{1}{8} \times P(S) = \frac{1}{10}$$
$$P(S) = \frac{4}{5}$$

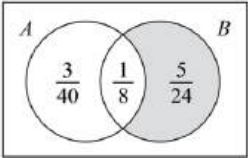
c. $P(\text{at least 1 not available}) = 1 - P(\text{all are available})$

◆◆ Mean mark (c) 34%.

$$= 1 - \left(\frac{4}{5}\right)^4$$
$$= \frac{369}{625}$$

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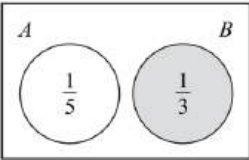
i. Sketch Venn diagram:



$$\therefore P(A \cap B) = P(B) - P(A \cap B)$$
$$= \frac{1}{3} - \frac{1}{8}$$
$$= \frac{5}{24}$$

◆◆ Mean mark 31%.
MARKER'S COMMENTS: Students who drew a Venn diagram or Karnaugh map were the most successful.

ii.



Mutually exclusive means $P(A \cap B) = 0$,

$$\therefore P(A \cap B) = \frac{1}{3}$$

◆◆ Mean mark 23%.